

SPECIAL PROJECT SPECIFICATION

ADD Section 301

SECTION 301- BASE FAILURE REPAIR

Description

301.01 This work consists of furnishing equipment, labor and materials to repair roadway base failures. This work also includes cutting asphalt, excavation of base failure areas, placement of geotextile fabric, and the furnishing, hauling, placement and compaction of rock to create underdrains as specified in the project plans and these specifications.

Material

301.02 Conform to the following Section and Subsections:

Geotextile

715.11.4

301.03 General. Elevate and protect sheets, panels, or rolls of geotextile filter with a waterproof and ultraviolet resistant cover if stored outdoors. Limit geotextile filter exposure to less than 10 days when using for a permanent installation.

In base failure repair areas that occur in hot-mix asphalt, paved travel ways, the asphalt pavement will be saw cut to the full depth of the asphalt. Joints shall be cut in a manner as to construct straight, neat longitudinal and transverse joints. Transverse joints shall be cut at a 90-degree angle to the centerline of the road and Longitudinal joints shall be cut parallel with the centerline of the road, unless otherwise directed or approved by the COR.

The size and approximate location of the base failures to be repaired are shown in the plans. Determine the final width, length, and depth of repair in the field, based on existing conditions and as approved by the COR.

Do not install drain material until the COR has accepted the final location and extents of excavation.

Do not allow soil or other foreign material to enter the drain systems.

All base failure areas shall be daylighted as shown on the plans. The bottom of excavation of the base failure area and the daylight channel shall be shaped to drain to the fill slope unless otherwise approved by the COR or shown on the plans. Areas of drainage channel outside of the paved roadway will be covered utilizing native material to match existing ground surrounding

the area, compacted to match density of surrounding area as verified by visual inspection. The outlet of the drainage channel shall remain open and be free of soil and deleterious material.

All areas excavated for base failure repair, within the paved road prism, shall be filled to within 3 inches of final grade by the end of each day's operation.

CONSTRUCTION REQUIREMENTS

301.04 Excavation of Base Failure Areas - Excavate designated area to required minimum depth limits Including full failure area plus drainage channel. If unsuitable material is encountered at this depth continue as directed by the COR. Dispose of excavated material in an approved waste site off Forest Property and in accordance with 207.06.3.2 of the Standard specifications.

Smooth the excavated surfaces by removing projections that may damage the geotextile. Replace geotextile filter damaged during installation.

Immediately upon completion of excavation, the bottom of the excavated area will be smoothed, shaped to drain, and compacted. Material will be compacted to approximate density of surrounding ground as verified by visual inspection.

Measure and document extents of excavation including average length, average width, and average depth. At a minimum, the length will be measured at each edge of the excavation parallel with the center line of the roadway. At a minimum, the width of the excavation will be measured at each edge of the excavation perpendicular to the centerline. Depth of the excavation will be verified to meet minimum depth required in the plans and will be total excavation will be calculated using plan depth dimensions. Measurements will be taken to the nearest 0.1 ft. The main base failure area and the drainage channel shall be measured as two separate areas and volumes of excavation within the two areas combined to determine quantity of material removed.

301.05 Underdrains For Base Failure Repair

Place fabric for subsurface drainage (meeting the requirements of AASHTO 288, Section 8, Class2) so that it is smooth, taut, and wrinkle free on the underlying surface. Allow enough excess fabric to completely encapsulate the stone layer within the base failure and drainage channel areas. Overlap a minimum of 24 inches at all seams. If longitudinal overlaps are required, do not place longitudinal overlaps below anticipated wheel loads. Hold the geosynthetic in place with pins, staples, or piles of cover material.

Place underdrain stone on the geosynthetic fabric. Do not operate equipment directly on the fabric. Spread the drain material, maintaining a 12-inch lift thickness.

Rock used to create underdrains shall be angular stone from a rock quarry that is hard, durable, free of organic and soil material, and resistant to weathering and water action. Do not use crushed river rock or rock with rounded surfaces.

Conform to the following:

- A) Durability index (course), AASHTO T210 50 min
- B) Gradation, 90-100% passing 8" sieve, 0-5% passing 3" sieve with 4" and 6" intermediate sizes represented.
- C) Los Angeles abrasion, AASHTO T96 50% max

Consolidate the placed material by using a smooth drum vibratory roller or other means approved by the COR. Material will be considered properly consolidated when material shows no sign of deformation upon repeat of compaction effort. Continue placing material in lifts no greater than 12 inches until proper elevation, as shown in the plans $\pm 1/2"$, is attained. Top of gabion rock shall be shaped to match plan road template $\pm 1/2"$.

Fold fabric for subsurface drainage to complete full encapsulation of the underdrain stone. Overlap a minimum of 24 inches at all seams. The outlet end of the drainage channel is to be left open with no fabric or cover. Shape to match fill slope in the area.

MEASUREMENT

301.07 Measurement of the 301 pay items listed in the bid schedule as follows:

ITEM 301.04 "Excavation of base failure areas" – calculate CY of material remove by measuring average length, average width, and plan depth as described in subsection 307.04.

ITEM 307.05 "Construction of Underdrains for Base Failure repairs" - calculate CY of material placed by determining the average length and average width of the material placed by taking measurements for length and width as described in subsection 307.04. Depth for calculation shall be as shown in the project plans.

Fabric for subsurface drainage will be considered incidental to this item. Stone utilized in construction of Underdrain will be considered incidental to this item.

PAYMENT

Payment for underdrains shall be made by the CY and calculated based on the length and width of excavation at a depth of 12" for the base failure area and drain. The main base failure area and the drainage channel shall be calculated as two separate areas and volumes combined to determine the pay quantity for the underdrain item.

The accepted quantities will be paid at the contract price per unit of measurement for the Section 301 pay items listed in the bid schedule. Payment will be full compensation for the work as described in this section.